

TITLE PAGE

- **Problem Statement ID – 03**
- **Problem Statement Title- Autonomous Production Choke Controller for a Single Naturally Flowing Oil Well**
- **Theme- Industrial Automation and Artificial Intelligence**
- **PS Category- Software**
- **Student Name – Sarah Thomas**
- **Student ID – sarahtthomas29@gmail.com**

IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed Explanation

Developed an AI-driven autonomous choke controller.

Predicts oil production and pressure using machine learning.

Selects the optimal choke opening through simulation.

Ensures safe well operation using pressure constraints.

How It Addresses the Problem

Maximizes oil production.

Maintains safe operating limits.

Eliminates manual choke adjustments.

Adapts to changing well conditions.

Innovation & Uniqueness

Combines predictive analytics with autonomous control.

Safety-aware decision-making for choke optimization.

Easily adaptable to real-time oilfield operations.

TECHNICAL APPROACH

Technologies Used

Python

Google Colab / Jupyter Notebook

Pandas, NumPy (Data Processing)

Matplotlib, Seaborn (Visualization)

Scikit-learn (Linear Regression Model)

Methodology

Load & preprocess dataset.

Perform EDA and step-test analysis.

Train predictive models for oil rate and pressures.

Generate candidate choke positions.

Predict outputs and validate safety constraints.

Select the optimal choke setting.

FEASIBILITY AND VIABILITY

Feasibility

Uses readily available Python-based ML libraries.

Requires only historical well data for training.

Can be integrated with real-time oilfield simulators.

Potential Challenges & Risks

Limited dataset may affect prediction accuracy.

Well conditions may vary beyond training data.

Real-world deployment requires continuous validation.

Mitigation Strategies

Retrain models with new operational data.

Incorporate real-time sensor feedback.

Validate controller decisions with safety constraints before execution.

ARTIFACTS

Code Implementation

Python-based autonomous choke controller.

Machine learning model for production prediction.

```
▶ def generate_candidates(current_choke):  
  
    start = max(MIN_CHOKE, current_choke - MAX_STEP)  
  
    end = min(MAX_CHOKE, current_choke + MAX_STEP)  
  
    return list(range(int(start), int(end)+1))
```

ARTIFACTS

Solution Prototype

End-to-end workflow from data analysis to autonomous choke selection.

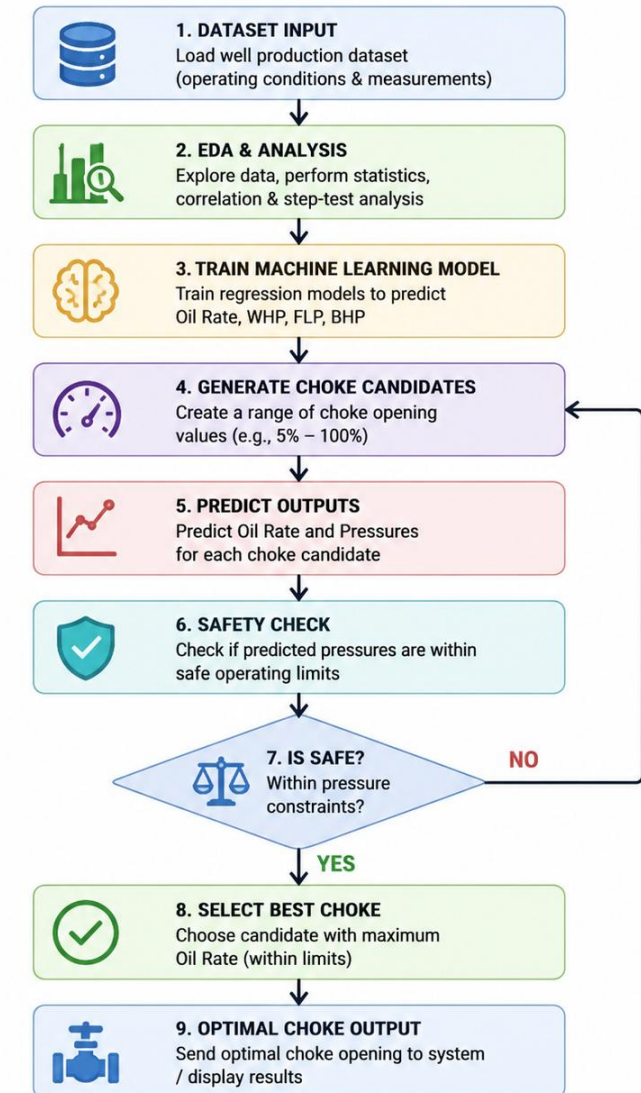
Safety-aware decision-making process.

Dashboard / Results

Oil production and pressure trend analysis.

Controller simulation results across scenarios.

AUTONOMOUS CHOKE CONTROLLER – WORKFLOW



RESEARCH AND REFERENCES

Honeywell Hackathon Problem Statement

Sample Dataset

Scikit-learn Documentation (Linear Regression)

Pandas Documentation (Data Processing)

Matplotlib & Seaborn Documentation (Visualization)

Python Official Documentation